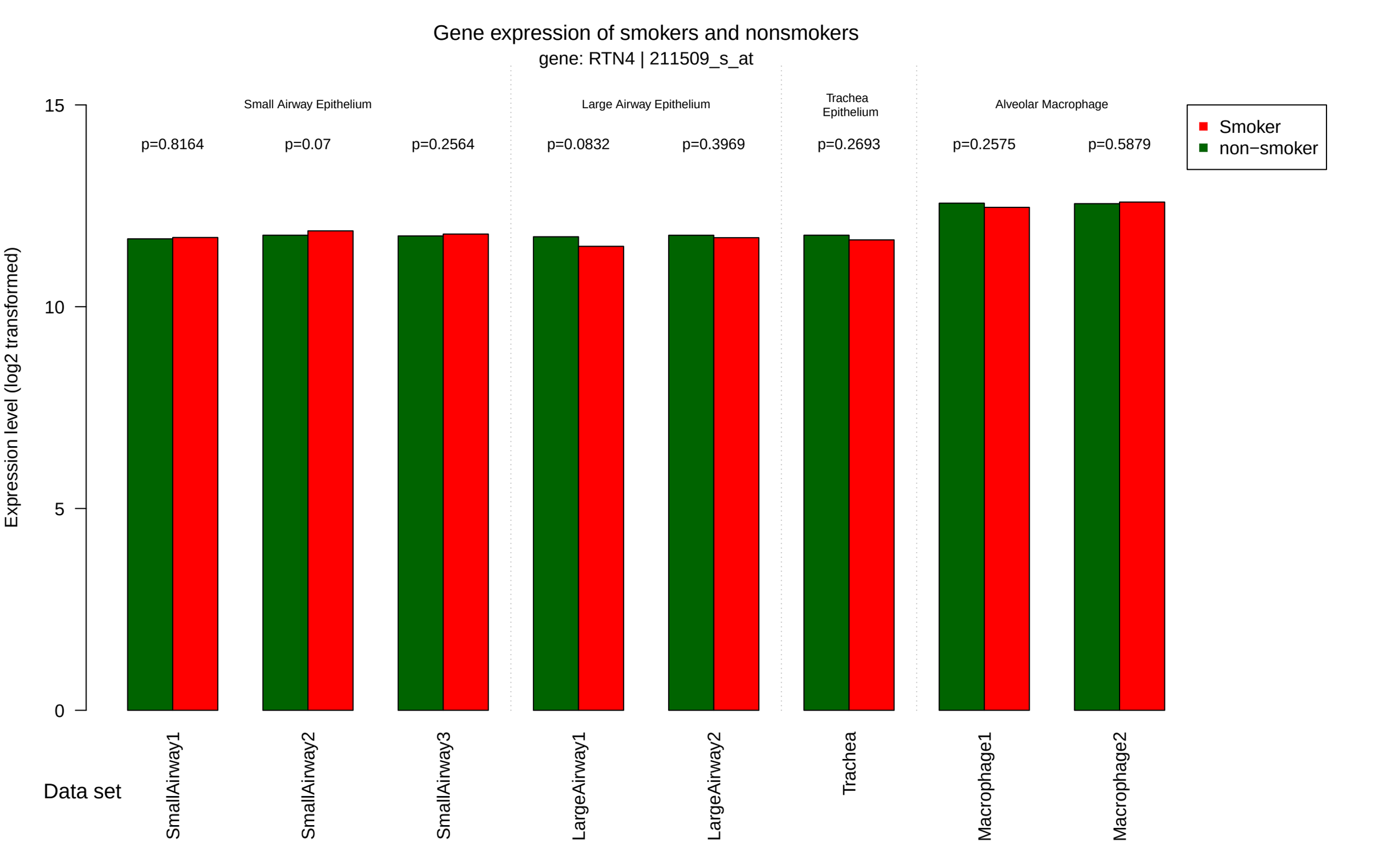
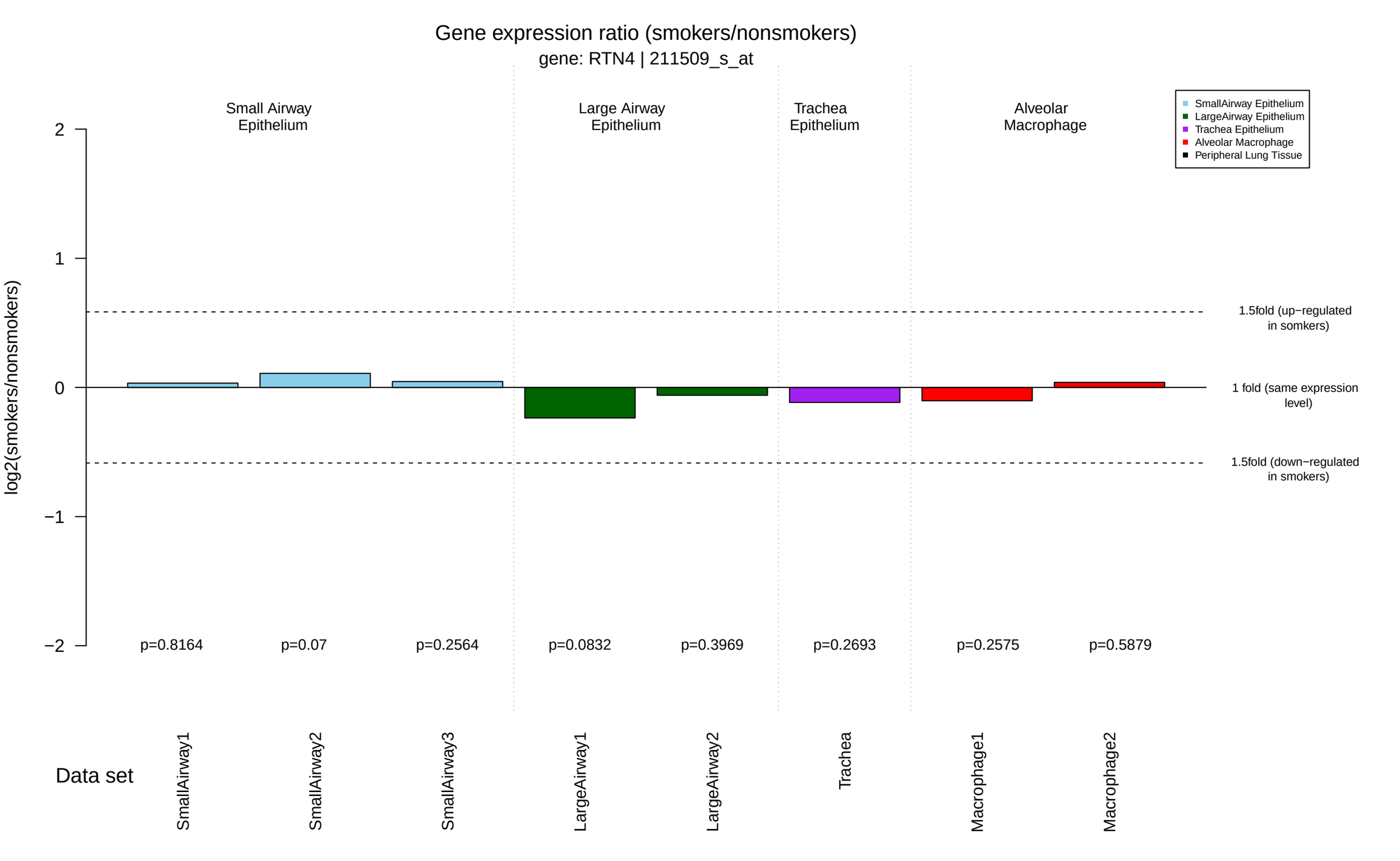
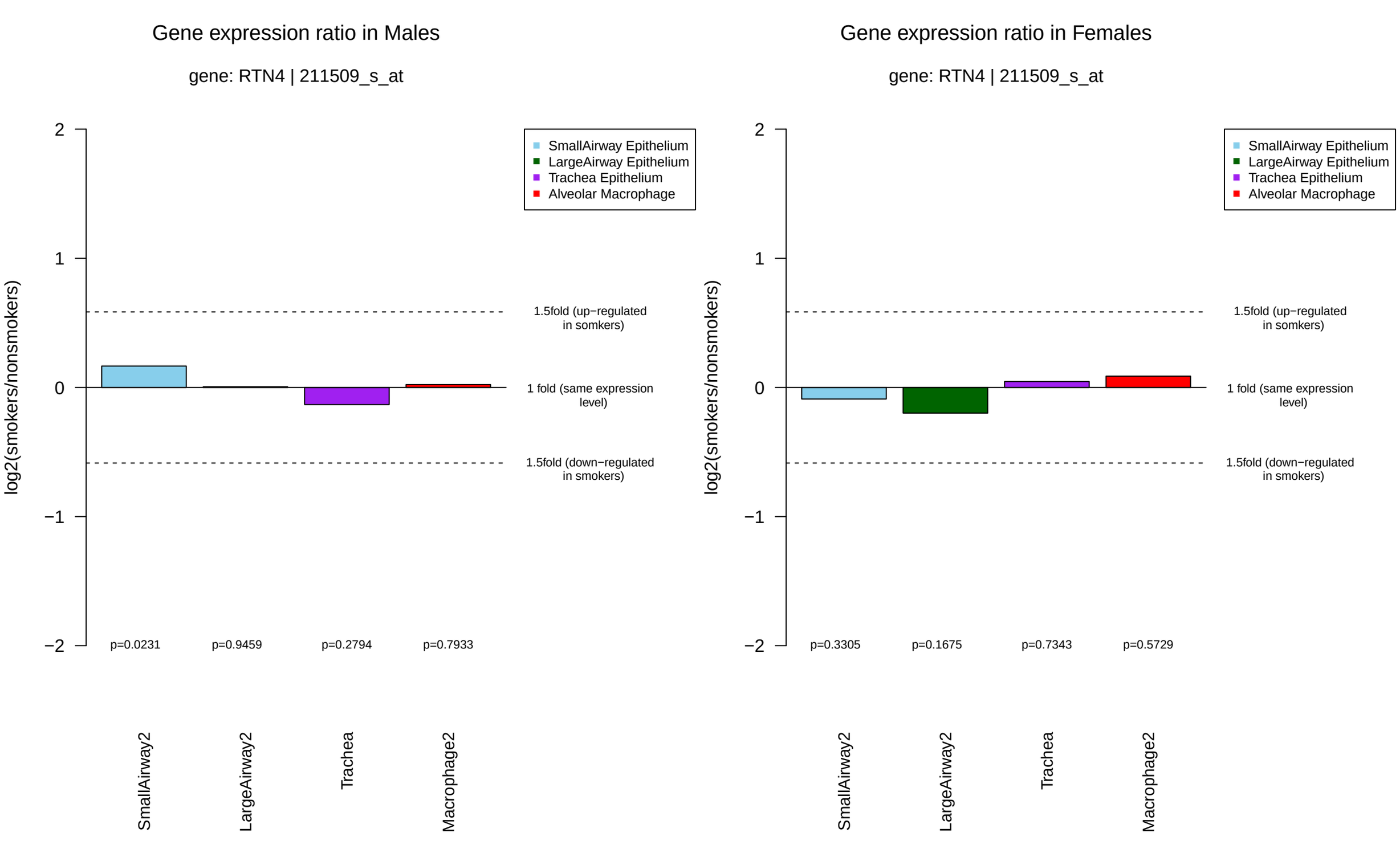
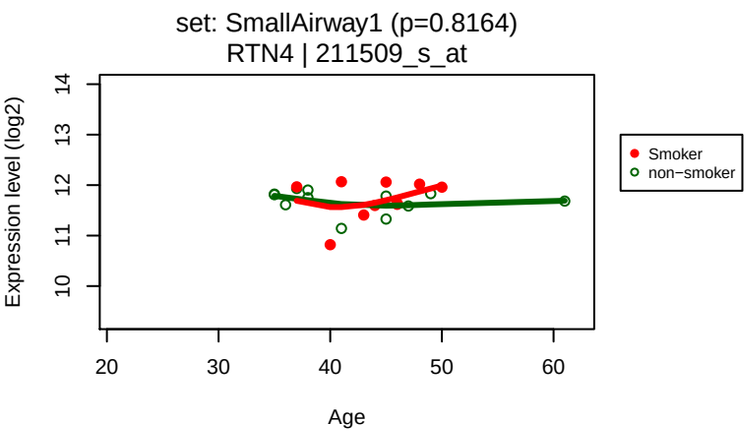
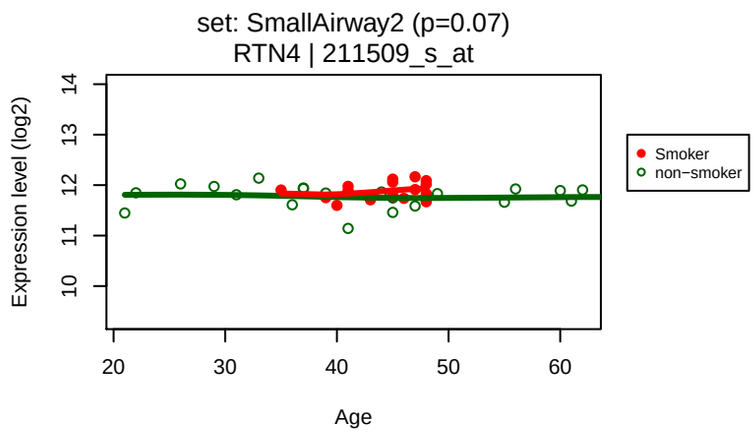
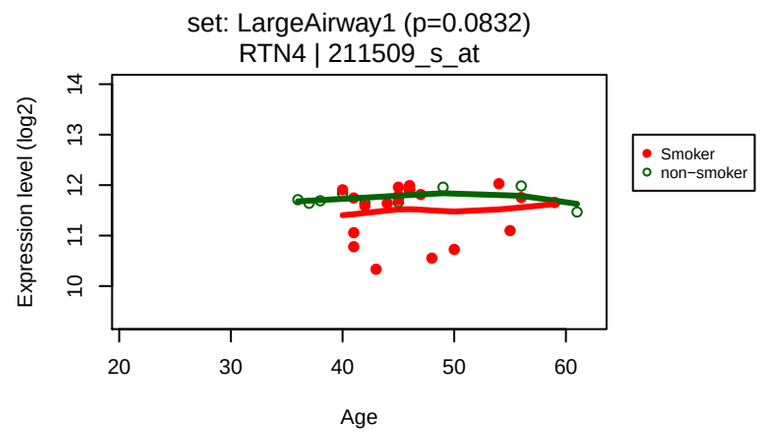

Gene expression of smokers and nonsmokers
gene: RTN4 | 211509_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8164
p=0.07
p=0.2564
p=0.0832
p=0.3969
p=0.2693
p=0.2575
p=0.5879
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

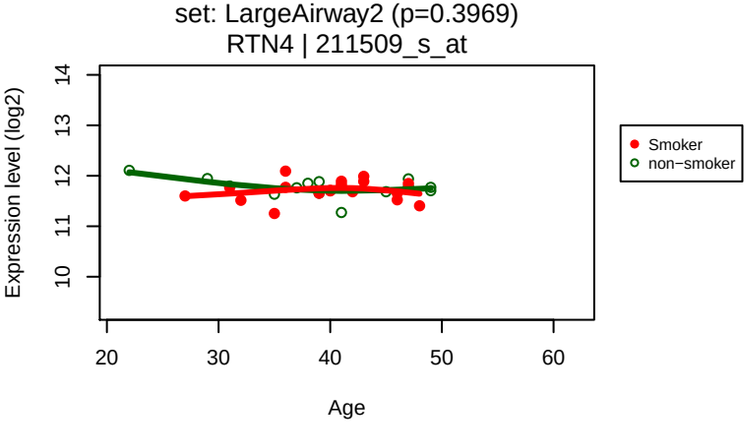

Gene expression ratio (smokers/nonsmokers)
gene: RTN4 | 211509_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8164
p=0.07
p=0.2564
p=0.0832
p=0.3969
p=0.2693
p=0.2575
p=0.5879
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

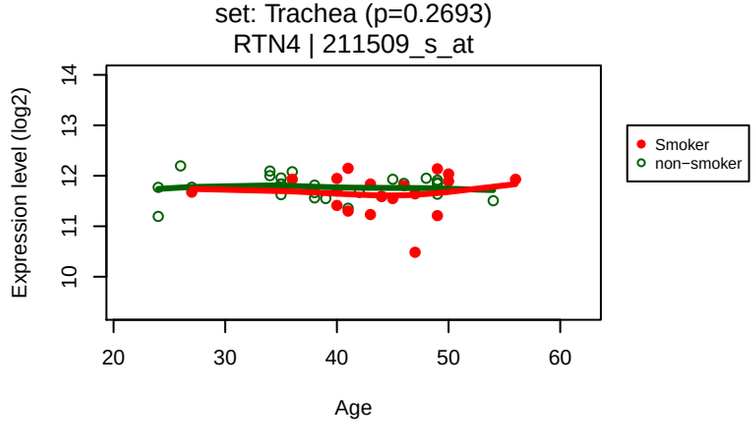

Gene expression ratio in Males
gene: RTN4 | 211509_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0231
p=0.9459
p=0.2794
p=0.7933
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN4 | 211509_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3305
p=0.1675
p=0.7343
p=0.5729
SmallAirway2
LargeAirway2
Trachea
Macrophage2

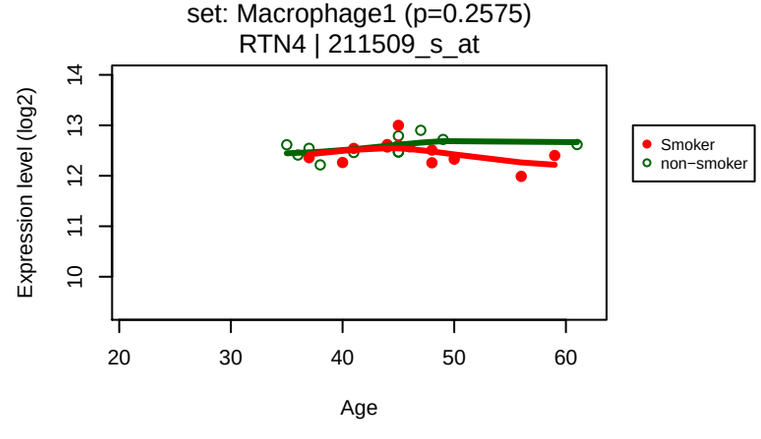

set: SmallAirway1 (p=0.8164)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker

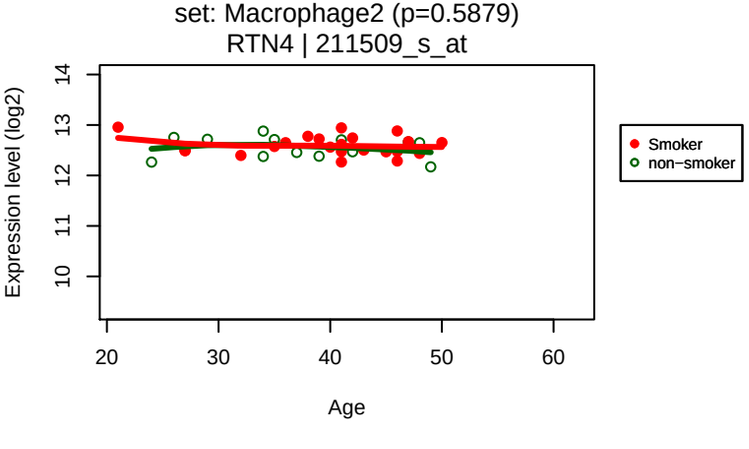

set: SmallAirway2 (p=0.07)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker

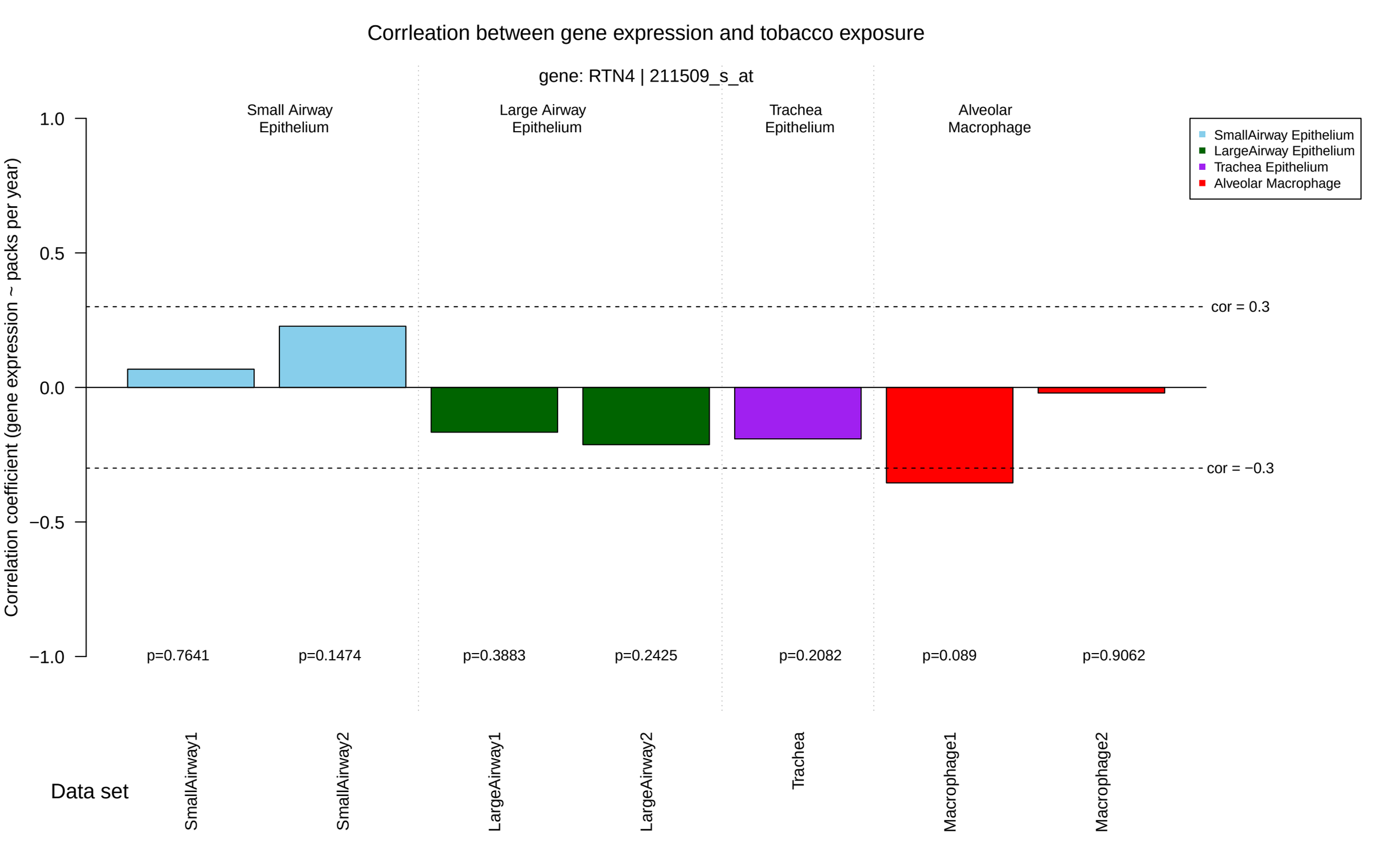

set: LargeAirway1 (p=0.0832)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3969)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2693)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2575)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5879)
RTN4 | 211509_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN4 | 211509_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7641
p=0.1474
p=0.3883
p=0.2425
p=0.2082
p=0.089
p=0.9062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2